Phenolic profiles and antioxidant properties of apple skin extracts.

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Lipid oxidation, especially the oxidation of polyunsaturated fatty acids, is a significant issue in the food industry impacting both food quality and health of consumers. Apple skin was investigated as a source of natural antioxidants. The phenolic compound composition and antioxidant properties of 21 selected apple genotypes were evaluated. The lipid stabilizing ability of the apple skin extracts was examined using an aqueous emulsion system of methyl linolenate. The total phenolic concentrations determined by high-performance liquid chromatography tandem mass spectrometry of methanolic extracts of skins of the apple genotypes varied from 150 to 700 mg/100 g DW. The antioxidant capacity measured by Folin-Ciocalteu (16.2 to 34.1 mg GAE/100 g DW), ferric reducing antioxidant power (1.3 to 3.3 g TE/100 g DW), oxygen radical absorbance capacity (5.2 to 14.2 g TE/100 g DW), and percent inhibition of oxidation of methyl linolenate (73.8% to 97.2%) varied among the apple genotypes. The apple skin extracts, specifically the crab apple varieties such as "Dolgo," were revealed to be effective inhibitors of oxidation of polyunsaturated fatty acid in a model system and thus can be considered as a potential source of natural food antioxidants.